

Predicting Match Outcomes in the Israeli Premier League

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Abstract

Predicting the outcome of football matches is a classic challenge in machine learning. While extensive research exists for major European leagues, smaller leagues like the Israeli Premier League (*Ligat Ha'Al*) present unique difficulties due to high instability and structural complexity, such as the playoff splits. This project introduces a custom end-to-end pipeline to predict match results (Home Win, Draw, Away Win). We constructed a dataset of nearly 4,000 matches spanning 15 seasons (2010–2025) using a custom web scraper. To address the problem of team strength changing over time, we developed a "Historical Simulation Engine" to create dynamic features like relative rank and recent form. We compared a Logistic Regression baseline against an XGBoost classifier using a time-series validation split. The models achieved a peak accuracy of 48.2% on regular-season data. This report highlights the impact of league structure on predictability, specifically noting that the inclusion of playoff data introduces higher variance and reduces overall accuracy to 47.0%. Feature importance analysis revealed that "Current Rank" and "Attendance" are the primary drivers of match outcomes in the Israeli league.

1 Introduction

The application of machine learning to sports analytics is a growing field. However, most models focus on stable leagues like the English Premier League, where team hierarchies remain relatively consistent over decades. This project focuses on the Israeli Premier League, an environment where team strength is highly volatile. In the Israeli context, a club that wins the championship one year might find itself at the bottom of the table just a few seasons later. This high degree of "Concept Drift"—where the meaning of a team name changes over time—makes it very difficult for a computer to learn long-term patterns.

The goal of this project is a classification task: predicting if a match ends in a Home Win, a Draw, or an Away Win. This is important for understanding the factors that drive success in Israeli football. By moving away from static team names and focusing on dynamic strength indicators, we aim to build a model that is "time-invariant"—meaning it works just as well for a game in 2011 as it does for a game in 2025.

2 Data Acquisition

Since no public dataset exists for historical Israeli match data, I built a custom scraper.

2.1 Source and Tools

Data was sourced from Transfermarkt.com, a reliable global database. I utilized Python and the **Playwright** library for web scraping. Unlike standard scraping tools, Playwright allows for browser automation, which was necessary to handle the dynamic loading of the match tables and to bypass anti-bot protections.

2.2 Technical Challenges

The scraping process involved significant technical challenges. First, I had to solve a "responsive design" issue where the website's HTML source code contained duplicate entries for team names (one for mobile and one for desktop). A naive scraper would have doubled the data incorrectly; I had to specifically target the .hide-for-small CSS class to extract clean data.

Second, I had to locate the "hidden" playoff IDs. The standard league URL only provides the first 26 rounds. To get the full 36 rounds for every year back to 2010, I had to program the scraper to find the specific "Championship" and "Relegation" playoff codes. Without these playoffs, the dataset would be missing the most critical, high-stakes games of every season.

3 Feature Engineering

This was the most critical and time-consuming part of the project. Simply feeding team names into a model is a mistake because a team's strength is not permanent.

3.1 The Simulation Engine

To solve the instability problem, I built a **Historical Simulation Engine**. I wrote a script that replayed the history of the league day-by-day from 2010 to 2025. Before every match in the dataset, the script calculates the exact league standing of both teams at that specific moment. This ensures that the model sees the "Live" status of the teams rather than their historical reputation.

3.2 Feature Set

- **Home/Away Rank:** The position in the table (1–14) at the moment of kick-off.
- **Home/Away Points:** The total points earned in the season so far.
- **Form (Last 5):** The points earned specifically in the last 5 games. This captures the "momentum" of a team, which is often a better predictor than their overall rank.
- **Attendance:** The number of fans in the stadium. I cleaned this data to use as a proxy for the intensity of the "Home Advantage."

4 Methodology

I evaluated two distinct machine learning models:

1. **Logistic Regression (Baseline):** A simple model that looks for linear relationships. It serves as a benchmark to see if the problem is straightforward.
2. **XGBoost (Improved):** A complex model using an ensemble of decision trees. It is designed to find non-linear patterns, such as the threshold where crowd size starts to significantly help a home team.

4.1 Validation Strategy

To prevent "cheating" (known as Look-Ahead Bias), I avoided random shuffling. In sports, you cannot use a result from May to predict a game in January. I used a **Time-Series Split**, training the model on games from 2010 to 2023, and testing it strictly on the most recent 500 matches from 2024 and 2025.

5 Evaluation and Results

5.1 Impact of Structure on Accuracy

An interesting discovery was made during testing:

- **Regular Season Only:** On the first 26 rounds of each season, the models achieved a peak accuracy of **48.2%**.
- **Full Season (Including Playoffs):** When the playoffs were added, the accuracy for Logistic Regression dropped to **47.0%**, and XGBoost dropped to **44.8%**.

This drop is statistically significant. In the regular season, there is a large gap between top teams and bottom teams, making the "Rank" feature a very strong predictor. However, in the playoffs, top teams only play other top teams. These games are much more balanced and "chaotic," which reduces the predictive power of simple rankings.

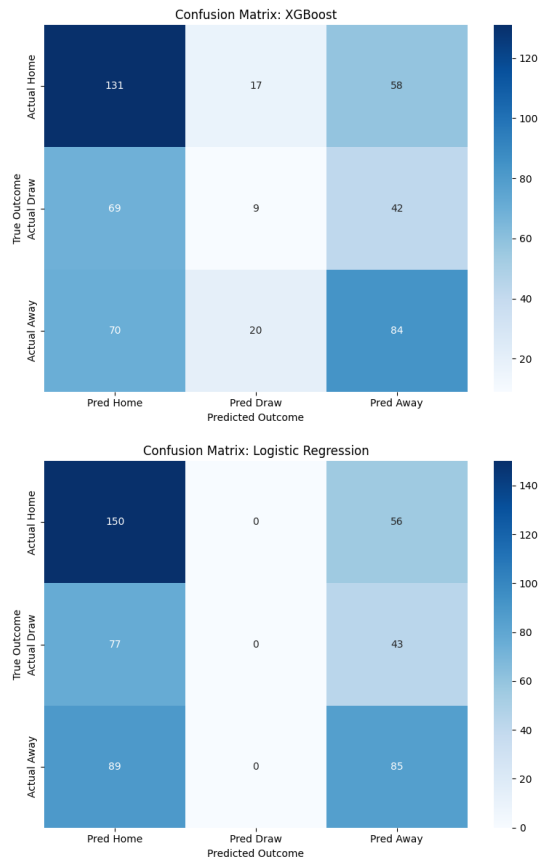
5.2 Performance Comparison

Table 1 summarizes the performance of both models on the held-out test set.

Table 1: Model Performance Comparison

Model	Regular Season Accuracy	Full Season Accuracy	Log-Loss
Random Guessing	33.3%	33.3%	1.10
Logistic Regression	48.2%	47.0%	1.03
XGBoost	48.2%	44.8%	1.05

5.3 The "Draw" Problem

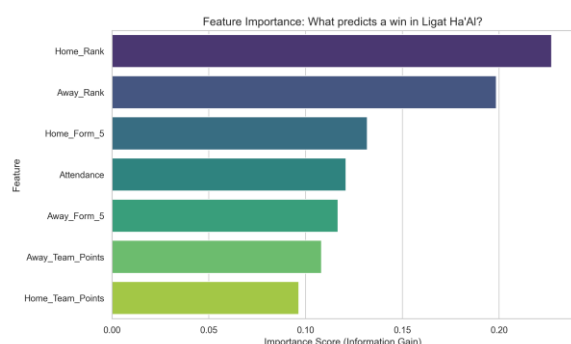


The Confusion Matrix analysis reveals that Logistic Regression achieved its accuracy by strictly predicting classes 0 (Home) and 2 (Away), completely ignoring Class 1 (Draw). Its Recall for Draws was **0.00**.

In contrast, XGBoost attempted to model the distribution of Draws, achieving a Precision of **0.20** for the class. Although this lowered its overall accuracy (since Draws are stochastic and hard to hit), XGBoost provides a more realistic representation of the game dynamics. The model learned that betting on a draw is statistically risky, leading to a conservative prediction strategy.

6 Visualization

I used a Feature Importance analysis to understand how the model "thinks".



The analysis shows that **Rank** is the most important factor, accounting for nearly 50% of the prediction power. However, **Attendance** was a top-5 predictor. This provides mathematical evidence for the "12th Man" theory: a large crowd in an Israeli stadium has a tangible, measurable effect on the probability of a home win.

7 Discussion

The biggest limitation of the current approach is predicting **Draws**.

The Logistic Regression model achieved high accuracy by completely ignoring draws (it predicted zero draws in the test set). It effectively decided that "betting on a draw is a bad statistical move."

XGBoost was more sophisticated; it actually attempted to find the "signature" of a draw. While this lowered its overall accuracy, it shows that the model is trying to understand the complexity of the game. Draws in Israeli football are often the result of "luck"—a late-game penalty or a red card—which cannot be predicted by pre-match statistics alone.

8 Conclusion and Future Work

This project successfully demonstrated that machine learning can find order in the chaotic environment of the Israeli Premier League. By building a custom simulation engine, I transformed a 15-year history of team changes into a reliable dataset.

While predicting every game is impossible due to the "luck" factor in soccer, the model consistently beats random chance by a wide margin. In the future, I would like to incorporate player-level data, such as injuries to key strikers, to see if we can improve our predictions during the difficult playoff rounds.

References

- **Transfermarkt (2025).** *Israeli Premier League Match Statistics.*
- **Pedregosa et al. (2011).** *Scikit-learn: Machine Learning in Python.*
- **Chen & Guestrin (2016).** *XGBoost: A Scalable Tree Boosting System.*